

Ekmat

A shared event ledger for co-lending dispute adjudication, built on DRUNIX

DRUNIX Hackathon | CHL-7007 | Problem Statement 4: Financial Inclusion

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Sources throughout: RBI Co-Lending Directions 2025 | IIBF Bank-NBFC Co-Lending study | CRISIL Ratings | Business Standard | github.com/npci/drunix

The RBI (Co-Lending Arrangements) Directions, 2025 (RBI/DOR/2025-26/139, effective 1 January 2026) created a reconciliation surface and nothing to reconcile it on. **Para 25:** each lender keeps the borrower's account individually. **Para 26:** all money routes through escrow. **Para 31:** each reports to credit bureaus separately, for the same loan. **Para 33:** both must hold the same SMA/NPA classification by end of the next working day. **The Directions name no lead lender and no system of record** — nothing says whose version of the loan's history is authoritative when the two diverge.

TODAY Bank's LMS → Bureau: *"Standard"* • **CONFLICT** • NBFC's LMS → Bureau: *"SMA-2"*

The borrower's file carries a dispute they did not cause and cannot trace.

WITH EKMAT Bank's LMS + NBFC's LMS → **shared event ledger** → Bureau: *one agreed position*

Both lenders keep their own books. What they share is the sequence those books derive from.

What Ekmat does

One agreed event sequence. Every servicing event is written once, co-signed by both lenders, and ordered. Neither can later assert a different sequence, because neither holds the log alone.

Convergence before submission. Para 33 becomes a state transition, not a file exchange. Submission reads from the agreed state, so a contradictory pair cannot leave the system.

Adjudicable disputes. Escrow appropriation order, interest apportionment, the date a payment landed: resolved against a record neither party authored alone.

Architecture

Four organisations on one DRUNIX channel per arrangement.

BANK	endorser, own books (Para 25)
NBFC	endorser, own books (Para 25)
ESCROW BANK	writes settlement events
REGULATOR	observer, no write path

Endorsement policy is the load-bearing decision: `AND(BankMSP.member, NBFCMSP.member)`. Neither lender can append to the shared sequence alone, enforced at chaincode rather than by convention.

Why DRUNIX. NPCI's open source ledger, an enhanced fork of Hyperledger Fabric. Permissioned MSP fits a contractual counterparty set; channels and private data collections keep the shared sequence separate from either party's wider book. Java chaincode is verified in DRUNIX v1.0.0 source.

The evidence

"I think everybody is struggling. Fundamentally, it means your underwriting standards should be aligned with those of the originator."
Prashant Kumar, MD and CEO, Yes Bank, May 2025

IIBF surveyed 16 banks and 20 NBFCs: partners run different loan management systems, take one to two quarters to integrate, and IRAC versus Ind-AS divergence produces conflicting positions.

CRISIL Ratings: NBFC co-lending AUM crossed **Rs 1.1 lakh crore** at 31 March 2025, growing **35 to 40 percent** annually. Executives at UGRO, Tyger Capital, Sa-Dhan, ICRA and Vivriti have raised inconsistent classification across partners on record.

Already built

Regulatory reporting pipeline, merged upstream. CB-ILD, built inside openMF during a paid global internship with the Mifos Initiative. Ingests financial records, validates each against a regulatory taxonomy, handles rejections, retries and disputes. **11 REST APIs**, 3-role access control, audit logging on every write, 11 migrations. **6 pull requests merged** into the production codebase, each reviewed by senior maintainers.

Append-only ledger primitives, public code. Arbitrator. Hash-chained log, immutable audit storage, idempotent writes keyed by request so a retry processes exactly once under concurrent load. **70 passing tests**, including 9 adversarial attacks covering concurrent duplicates, replay and gate bypass. All 9 blocked, running on every commit.

Build plan

0	Days 1–2	Prove Java chaincode on DRUNIX, or pivot to Go
1	Days 3–5	Four orgs, channel, event schema
2	Days 6–9	Co-signed exactly-once concurrency append, under
3	Days 10–13	Para 33 convergence state machine
4	Days 14–16	Bureau submission gated on agreed state
5	Days 17–20	Dispute adjudication, demonstration

Phase 2 is the critical correctness milestone, scheduled early for that reason. Phases 0 to 4 are the minimum viable path; Phase 5 absorbs slippage.

The objection this must answer

"Why not a signed append-only log in Postgres?" Fair, and worth raising rather than dodging.

The answer is custody, not cryptography. A signed log in one party's database is still that party's log. Bank and NBFC have opposed incentives on NPA timing and collection sequencing, which is exactly why neither should host the record the other is bound by.

On why consortium ledgers died before. we.trade, TradeLens, Marco Polo and Contour failed on getting many counterparties onto one platform, not on technology. Co-lending is two parties with a contract and a regulatory duty to agree.

What the ledger buys: non-repudiable event ordering. When the two submit conflicting DPD, the dispute is whose timestamp was right.

Stated risks

Java chaincode on DRUNIX is undocumented by NPCI; Phase 0 retires this in two days, with a Go fallback. No live counterparty is available, so the demonstration runs two simulated lender organisations against synthetic data. No production deployment is claimed.